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Studierichting: Informatica
Oefeningenreeks 4A nr. 36.14

$$\int \frac{4dx}{2x^2 + 7} = \frac{4}{7} \int \frac{dx}{\left(\sqrt{\frac{2}{7}}x\right)^2 + 1}$$

$$\text{Neem } u = \sqrt{\frac{2}{7}}x$$

$$du = \sqrt{\frac{2}{7}}dx \Rightarrow dx = \sqrt{\frac{7}{2}}du$$

$$\begin{aligned} \text{Dan } \int \frac{4dx}{2x^2 + 7} &= \frac{4\sqrt{7}}{7\sqrt{2}} \int \frac{du}{u^2 + 1} \\ &= \frac{2\sqrt{14}}{7} \operatorname{Bgtan} \left(\sqrt{\frac{2}{7}}x \right) + c \text{ met } c \in \mathbb{R} \end{aligned}$$

References